Program Assignments – While Loops. Develop an IPO for each of the problems below. Place the IPO into your repository. Then write code for each problem and place those files (py) into your repository. Paste the link to your repository into the Assignment Completion Link on Blackboard.

1. Develop flowgorithm and code this this problem. Display the odd numbers starting at 1 and ending with 25. Use a while loop structure for this problem.

Input	Process	output
	Set start = 1	
Start = 1	While starting _< 25	Display odd numbers
Stop = 25	Display start	1,3,5, …, 25
Increment = 2	Start = start + 2	

2. Allow the user to enter a start value, stop value and increment value from the keyboard. Display all the numbers from the start value to stop value using the increment value as you proceed. Use a while loop structure for this problem.

input	process	Output
Start value		
Stop value	Get start, stop, increment	Display sequence of numbers
Increment value	While start _< stop	Numbers from start to stop
	Display start	
	Start = start + increment	

3. Prompt the user on whether they want to do this program **(just before the while loop**). "Yes" entry means they want to continue. Any other response indicates they will stop the program. This response is the loop control variable. If the user answers "Yes "then go into the while loop.

   Once in the while loop. You are to prompt the user for their last name and two exam scores. Compute the average exam score. Display last name and average. After the loop, display a count of the number of students who entered data.

   Finally, the **last statements within the while loop** will ask the user if they want to do this loop again. In other words the user needs to be prompted again. The reason is that the end of the loop takes execution to the while condition to be evaluated again. It can not take us to the first few lines of code that prompt the user for the first time. That code is out of the loop. Therefore, we need a second prompt at the bottom, inside the loop.

input	process	Output

Response (yes/no)	Initial prompt before loop	
Lname	While response = yes	Lname and avg
Exam1, exam2	Compute average = (exam1 + exam2) / 2	Number of students
	Display lname and avg	
	Count = count + 1	
	Ask user again (sentinel)	

4. Prompt the user on whether they want to do this program **(just before the while loop**). Yes means they want to continue. Any other response indicates they will stop the program. This response is the loop control variable. If the user answers Yes then go into the while loop.

   Once in the while loop. You are to prompt the user for employee last name, hours worked and rate of pay. Compute gross pay. Give the employee time and a half for hours worked over 40. Sum the gross pay and count the number of employees.

   For each employee display their last name and gross pay.

   After the loop (all data entered) display the sum of all the gross pays, and count of the number of employees. Compute and display the average pay.

   Finally, the **last statements within the while loop** will ask the user if they want to do this loop again. In other words the user needs to be prompted again. The reason is that the end of the loop takes execution to the while condition to be evaluated again. It can not take us to the first few lines of code that prompt the user for the first time. That code is out of the loop. Therefore, we need a second prompt at the bottom, inside the loop.

input	process	Output
	Initial prompt before loop	
Response (yes/no)	While response == yes	Lname and gross pay
Lname	If hours > 40 ->overtime	Total gross pay
Hours worked	Gross = (40 x rate) + (hours-40 x rate x 1.5)	Number of employees
Rate of pay	Else gross = hours x rate	Average pay
	Sum gross pay	
	Count employees	
	Ask again (sentinel)	

5. Prompt the user on whether they want to do this program **(just before the while loop**). Yes means they want to continue. Any other response indicates they will stop the program. This response is the loop control variable. If the user answers Yes then go into the while loop.

   Once in the while loop. You are to prompt the user for quantity and price of an item. Compute extended price (quantity times price of an item. If the extended price is greater than 10000.00 compute a discount of 25%. All other orders get a 10% discount. For each order display extended price, discount amount (extended price x discount percent), total (extended price – discount amount).

   For each order sum the discount amount.

   After the loop (all data entered) display the sum of all the discounts.

   Finally, the **last statements within the while loop** will ask the user if they want to do this loop again. In other words the user needs to be prompted again. The reason is that the end of the loop takes execution to the while condition to be evaluated again. It can not take us to the first few lines of code that prompt the user for the first time. That code is out of the loop. Therefore, we need a second prompt at the bottom, inside the loop.

input	process	output
Response (yes/no)	Initial prompt before loop	Extended price
Quantity	While response = yes	Discounted amount
price	Extended = quantity x price	Total price
	If extended > 10000 -> 25% discounts	Sum of discounts
	Else -> 10% discount	
	Discount = extended x rate	
	Total = extended – discount	
	Sum discounts	
	Ask again (sentinel)	